10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 9 — Tangent and Angles of a Circle

Unit 9 | Review + MCQs | New Book (PECTAA)

Review Exercise 9 (Solved)

The review covers the whole unit: tangents and their lengths, contact of two circles, angles at the centre and at the circumference, cyclic quadrilaterals, arc length and area of a sector.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) Tangent is a line that touches the circumference of the circle at:

(a) many points (b) three points
(c) two points (d) one point

Answer: (d) one point — a tangent touches the circle at exactly one point

(ii) _____ tangents can be drawn to a circle from a point outside the circle.

(a) one (b) two
(c) three (d) many

Answer: (b) two — from an outside point exactly two tangents can be drawn

(iii) _____ tangents can be drawn to a circle from centre of the circle.

(a) zero (b) one
(c) two (d) three

Answer: (a) zero — every line through the centre cuts the circle twice, so none is a tangent

(iv) In circle, radius and tangent are:

(a) parallel (b) equal
(c) perpendicular (d) zero

Answer: (c) perpendicular — the radius to the point of contact is perpendicular to the tangent

(v) An angle in a segment greater than a semicircle is _____ angle:

(a) an acute (b) right
(c) an obtuse (d) straight

Answer: (a) an acute — the bigger the segment, the smaller the angle

(vi) The angle subtended by the arc at the centre of the circle is called:

(a) acute angle (b) central angle
(c) right angle (d) complete angle

Answer: (b) central angle — its vertex is the centre of the circle

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(vii) What is the measure of an angle inscribed in a semi-circle?

(a) 45° (b) 60°
(c) 90° (d) 120°

Answer: (c) 90° — angle in a semi-circle is a right angle

(viii) Any two angles in the same segment of the circle are:

(a) supplementary (b) complementary
(c) equal (d) zero

Answer: (c) equal — angles in the same segment are equal

(ix) Area of the sector of the circle = _____

(a) ½ rθ (b) 3/2 $r^2\theta$
(c) ½ $r^2\theta$ (d) 3/2 rθ

Answer: (c) ½ $r^2\theta$ — sector area is half of r^2 times the angle in radians

(x) If r = 6 cm and θ = 2 radians, then arc length is:

(a) 12 cm (b) 8 cm
(c) 6 cm (d) 3 cm

Answer: (a) 12 cm — $l = r\theta = 6 \times 2 = 12$ cm

2. In the adjoining figure, find the radius of the circle.

In the figure PQ is a tangent of length 24 cm from the outside point Q, and OQ = 25 cm.

The radius OP is perpendicular to the tangent PQ

$OP^2 = OQ^2 - PQ^2$

$OP^2 = 25^2 - 24^2 = 625 - 576 = 49$

OP = 7 cm

Answer: Radius of the circle = 7 cm

3. Find the angles a, b, c and d in the given figure, if O is the centre of the circle.

In the figure P, Q, R, S lie on the circle with centre O, ∠PSQ = 42° and ∠QSR = 72°.

The line SQ passes through the centre O, so SQ is a diameter

The angle in a semi-circle is 90°, so

a = ∠SPQ = 90° b = ∠SRQ = 90°

In triangle PSQ: c = 180° − 90° − 42°

c = 48°

In triangle SRQ: d = 180° − 90° − 72°

d = 18°

Answer: a = 90°, b = 90°, c = 48°, d = 18°

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4. Find the angles a, b and c, if ML is a tangent line and O is the centre of the circle.

In the figure the tangent ML touches the circle at L, M, C and O lie on one straight line, and ∠LMC = 22°.

The radius OL is perpendicular to the tangent ML

In the right triangle OLM: $a = \angle LOM = 90° - 22° = 68°$

OL = OC (radii), so triangle OLC is isosceles

$c = \angle OCL = \frac{180° - 68°}{2} = 56°$

$\angle LCM = 180° - 56° = 124°$

In triangle LMC: $b = 180° - 124° - 22°$

$b = 34°$

Answer: a = 68°, b = 34°, c = 56°

5. Find the angles marked in the given figure, where O is the centre of the circle.

In the figure S and R lie on the circle, ∠SOR = 100°, a = ∠OSR and b = ∠ORS.

OS and OR are radii, so triangle SOR is isosceles

$a + b + 100° = 180°$

$2a = 80°$ (because a = b)

$a = b = 40°$

Answer: a = 40°, b = 40°

6. Two round dining tables are arranged to touch at their edges. The radii of the two round dining tables are 10 cm and 12 cm. What is the distance between their centres if they touch externally?

$d = r_1 + r_2 = 10 + 12$

$d = 22$ cm

Answer: Distance between the centres = 22 cm

7. A circular mosaic pattern is laid on the floor beneath a modern pyramid structure. Two tiles are placed such that they both connect to the same chord EF and reach two points G and H on the arc above EF. If m∠EGF = 65°, find m∠EHF.

G and H lie on the same side of the chord EF

Angles in the same segment of a circle are equal

Answer: m∠EHF = 65°

8. A triangular frame within a semicircular gate joins the ends of the diameter to the peak. What is the angle formed at the peak?

The peak lies on the semi-circle and the base is the diameter

The angle in a semi-circle is a right angle

Answer: The angle at the peak = 90°

9. A circular birthday cake has a radius of 15 cm. A slice has an arc length of 10 cm. What is the area of this slice?

$l = r\theta$ gives $\theta = \frac{l}{r} = \frac{10}{15} = 0.6667$ radian

Area $= \frac{1}{2} rl = \frac{1}{2} \times 15 \times 10$

$= 75\ cm^2$

Answer: Area of the slice = 75 cm²

10. A sector cut from a circle of radius 5 cm has a perimeter of 16 cm. Find area of this sector.

Perimeter of sector $= 2r + l = 16$

$2(5) + l = 16 \implies l = 6$ cm

Area $= \frac{1}{2} rl = \frac{1}{2} \times 5 \times 6$

$= 15\ cm^2$

Answer: Area of the sector = 15 cm²

Quick Revision — Unit 9 at a Glance

Tangent ⊥ radius at the point of contact; two tangents from an outside point are equal.

External contact: $d = r_1 + r_2$. Internal contact: $d = R - r$.

Angle at the centre = 2 × angle at the circumference.

Angle in a semi-circle = 90°; angles in the same segment are equal.

Opposite angles of a cyclic quadrilateral add to 180°.

arc = $r\theta$, area of sector = ½ $r^2\theta$ = ½ rl, with θ in radians.